Machine Learning

Gradient Descent, Part 1

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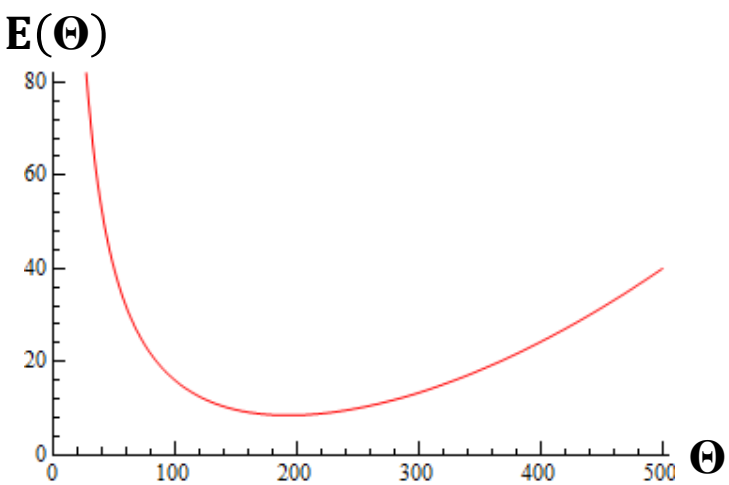
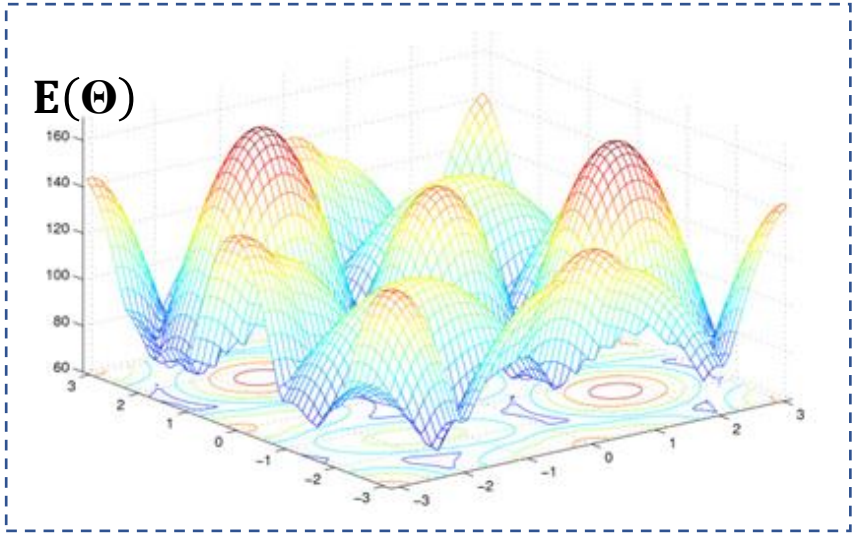
Model Training

1. **Define a cost (or, loss, error) function $\mathbf{E}(\boldsymbol{\Theta})$**
   - Difference between the model's predictions and actual values
2. **Solve the optimization problem**
   - Getting optimal $\boldsymbol{\Theta}^*$ that minimizes $\mathbf{E}(\boldsymbol{\Theta})$

- **Cases**



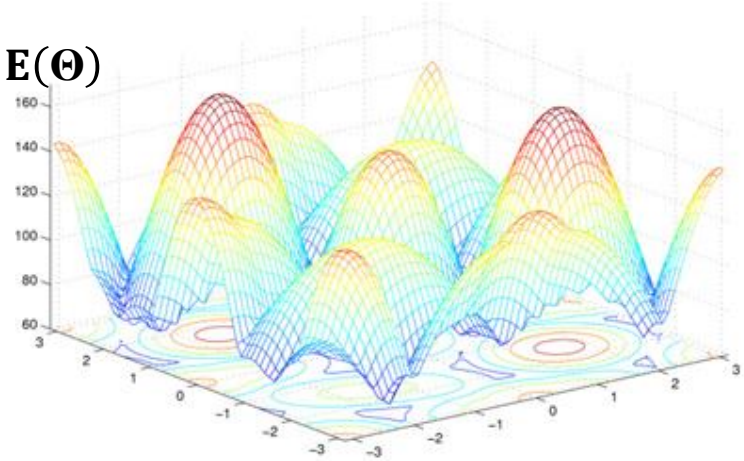
<E is convex>



<E is non-convex>

Numerical Optimization: Gradient Descent

- **If the cost function E is complex:**
  - There are multiple minimum points
    - Local minima
    - Global minima
  - We cannot find a minimum point directly
  - We cannot distinguish either it is global minima or local minima



- **Gradient descent**
  - https://en.wikipedia.org/wiki/Gradient
  - The **gradient** of a function **E** at a point **p** is the **vector** whose elements are the **partial derivatives** of **E** at **p**
  - The **gradient** can be interpreted as the "direction in which the function **E** **increases** **most quickly**"

Appendix: Derivative of a Scalar by a Vector

- **Let a scalar y and a vector $\mathbf{x} = [x_1 \quad x_2 \quad \cdots \quad x_n]$**
  - The derivative of y by x is:

$$\frac{\partial y}{\partial \mathbf{x}} = \left[\frac{\partial y}{\partial x_1} \quad \frac{\partial y}{\partial x_2} \quad \cdots \quad \frac{\partial y}{\partial x_n} \right]$$

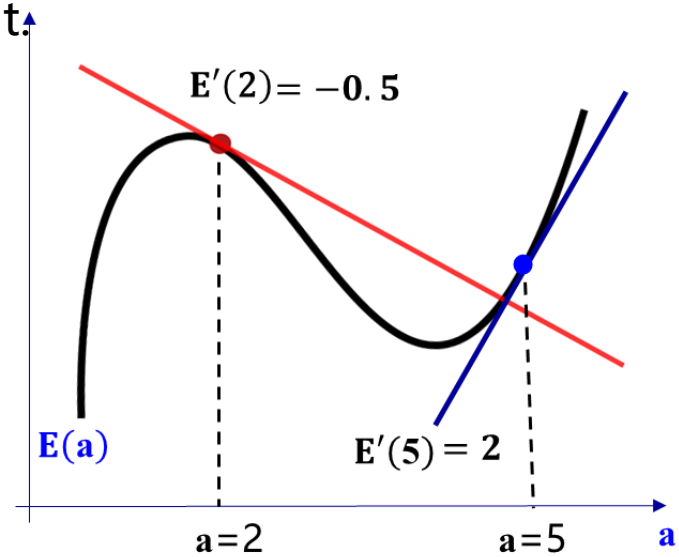
- **The gradient of a function (scalar) f at a point p is the vector whose elements are the partial derivatives of f at p**

$$\nabla f(p) = \left[\frac{\partial f}{\partial x_1}(p) \quad \cdots \quad \frac{\partial f}{\partial x_n}(p) \right]$$

Gradient Descent

Derivative

- The derivative of a function measures the "**instantaneous rate of change (순간 변화량)**" of the function at a specific point.
- $\mathbf{E'(2) = -0.5}$
  - E is **decreasing** at $\mathbf{a}$=2
  - Its instantaneous rate of decreasing is 0.5
- $\mathbf{E'(5) = 2}$
  - E is **increasing** at $\mathbf{a}$=5
  - Its instantaneous rate of increasing is 2



Partial derivative

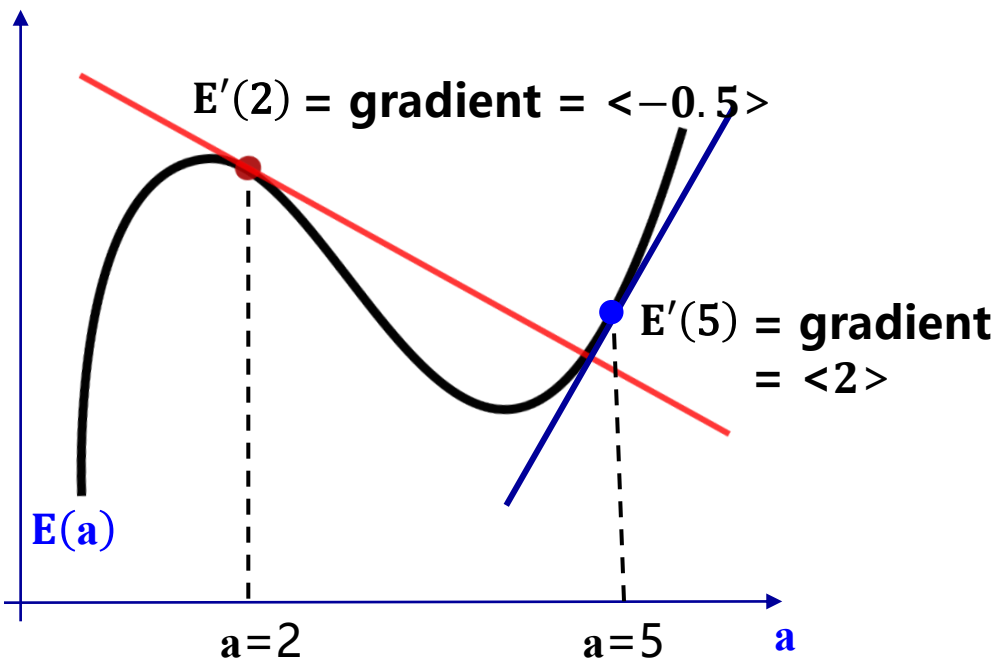
- The **partial** derivative of a function measures the **instantaneous rate of change** of the function at a specific point **with respect to one of parameters.**
  - $\frac{\partial E}{\partial a}$, $\frac{\partial E}{\partial b}$, $\frac{\partial E}{\partial c}$, $\frac{\partial E}{\partial d}$, if E is a function parameterized by *a, b, c,* and *d*
  - It implies the impact of each parameter on increase/decrease of the function

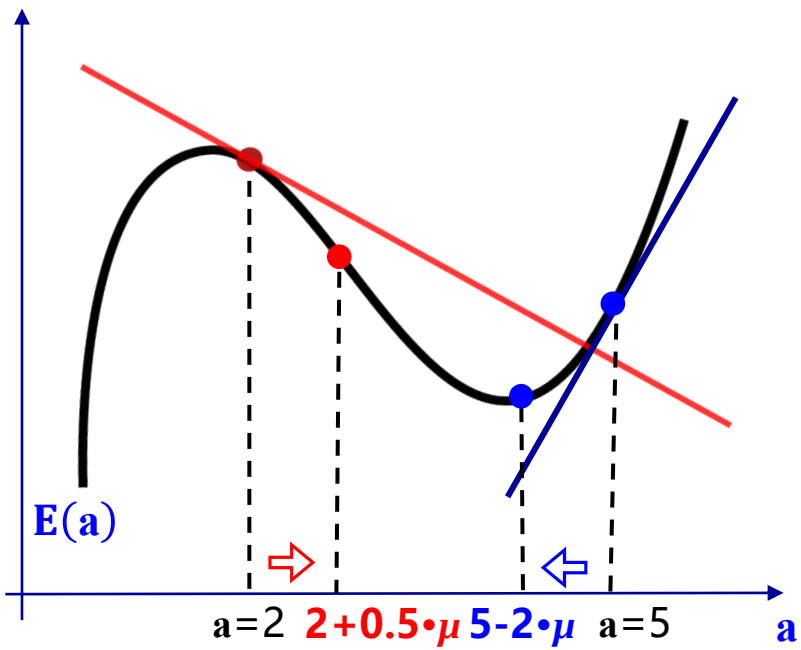
Gradient Descent

- The **gradient** at a point **p** is the **vector** pointing the direction in which the function **E is increasing most quickly**

$$< \frac{\partial E}{\partial a} \quad \frac{\partial E}{\partial b} \quad \frac{\partial E}{\partial c} \quad \frac{\partial E}{\partial d} >$$

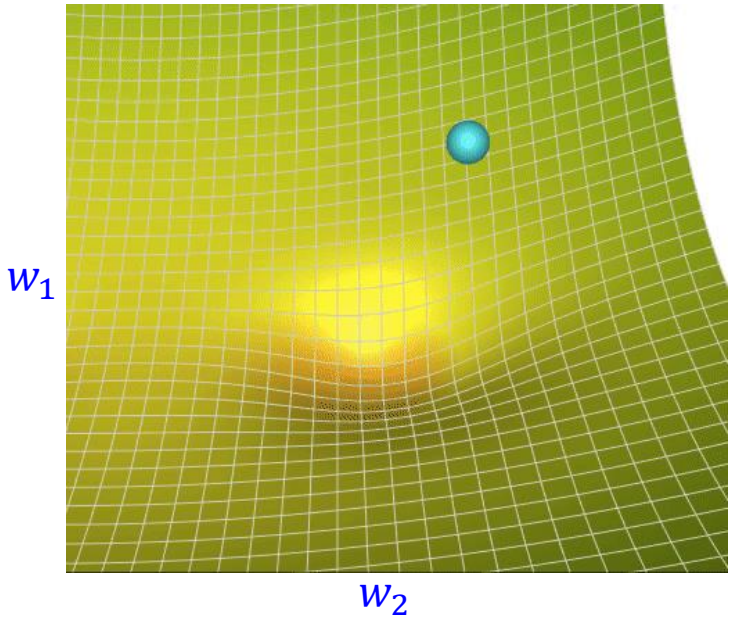
- **Gradient descent** is an iterative optimization algorithm for finding a local minimum of a function by **taking steps to the opposite direction of the gradient** at p, with a step size μ (or learning rate, typically less than 0.1).

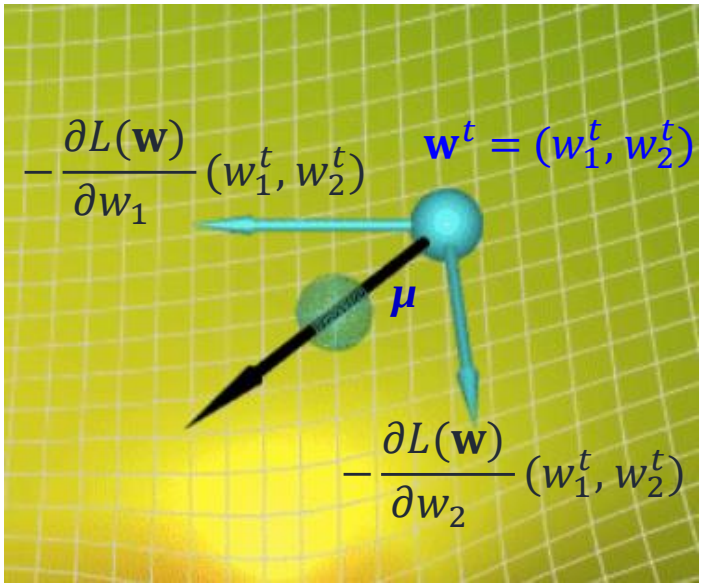




Gradient Descent

❑ 2D case



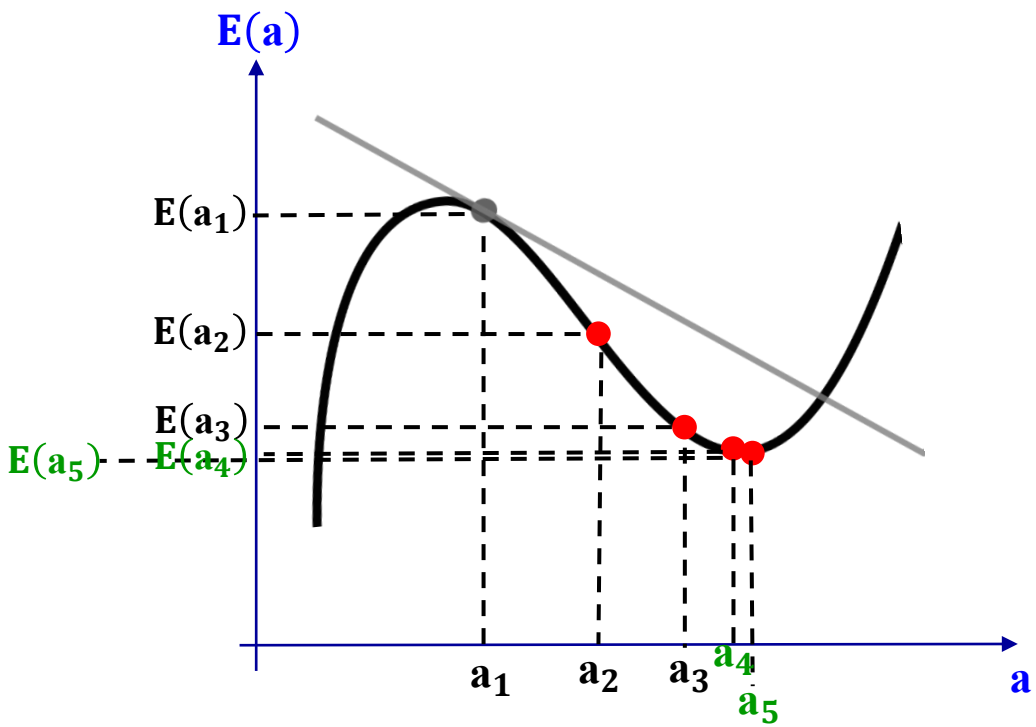


1. Initialize $\mathbf{w}^1$ randomly ($\mathbf{w}^1 = (w_1^1, w_2^1)$)
2. Compute **gradient** at $\mathbf{w}^1$, $<\frac{\partial L(\mathbf{w})}{\partial w_1}(w_1^1, w_2^1),\ \frac{\partial L(\mathbf{w})}{\partial w_2}(w_1^1, w_2^1)>$
3. Move from $\mathbf{w}^1$ to $\mathbf{w}^2$ towards **opposite direction** of gradient with a step size μ
   - $\mathbf{w}^2 = (w_1^1 - \mu \cdot \frac{\partial L(\mathbf{w})}{\partial w_1}(w_1^1, w_2^1),\quad w_2^1 - \mu \cdot \frac{\partial L(\mathbf{w})}{\partial w_2}(w_1^1, w_2^1))$
4. Repeat steps 2 and 3 until $L(\mathbf{w})$ converges

Gradient Descent

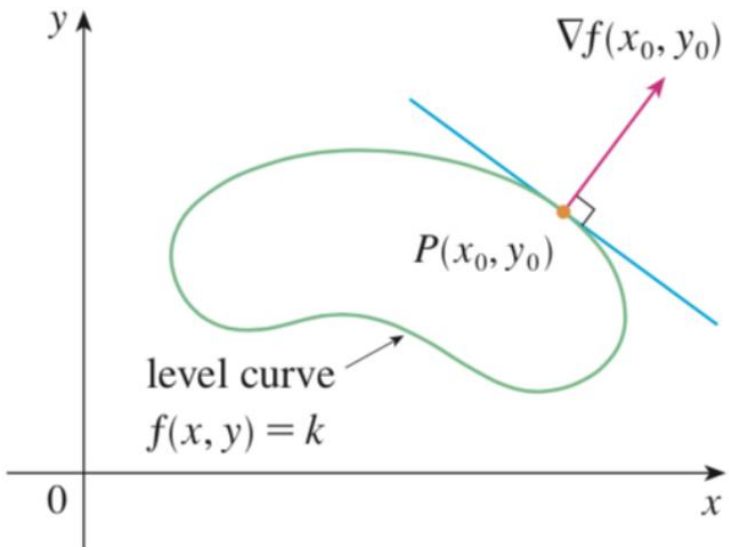
When to stop gradient descent

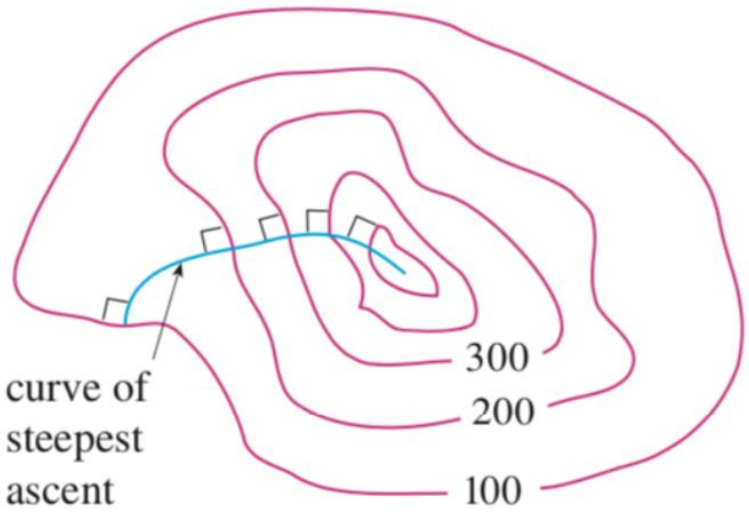
- Until the cost function **E** converges
  - Let δ be a convergence threshold, then we decide the convergence of E by:
  - If $\mathbf{E}(\mathbf{a}_{n-1}) - \mathbf{E}(\mathbf{a}_n) < \delta$
  - δ can be very small number like 0.001



Next Topic

- The **gradient** at a point **p** is the **vector** pointing the direction in which the function **E is increasing** **most quickly**





Thank You